

PHASE 2 — Strategy Application (In-Memory, no API calls)

Strategy Comparison

Strategy	Query Routing	Uniform Quota
Naive Retriever	✗	✗
QRe	✓	✗
UMS-Retriever	✗	✓
QRe & UMS-Retriever	✓	✓

$\varphi(\hat{y}_q) \rightarrow A_q$

S1 — Naive Retriever

Full Pool $\mathcal{P}_q$
all 3 sources · unsorted

Sort by cosine distance ↑

Top-k docs

No source filtering · risk of corpus dominance

S2 — QRe (Query Rerouting)

Filter $\mathcal{P}_q$ by $\hat{y}_q$
 $\mathcal{P} = \{d \in \mathcal{P}_q \mid \text{source}(d) \in A\}$

→ $A = \{G, I\}$
if newspapers only

→ $A = \{G, I, D\}$
if needs Débats

Sort filtered pool by distance ↑

Top-k docs

Source-filtered · no quota

S3 — UMS-Retriever

Compute per-source quota
 $n_{base} = \lfloor k / |C| \rfloor, r = k \bmod |C|$

Sort collections by min-distance ↑
distribute r extra slots to closest sources

Pick top q_c docs per collection c
 $\mathcal{P}_q = \{\mathcal{P}_x, \mathcal{P}_x, \mathcal{P}_x\}$
 $\mathcal{P}_{q,k,S} = \mathcal{P}_{q,k}[1:n_{base}+r]$ for S

Global re-sort by distance ↑

Top-k docs

Balanced coverage · no routing filter

S4 — QRe & UMS-Retriever Full strategy

Filter by routing: $\mathcal{P} = \{d \mid \text{source} \in A\}$

Quota over A (not C)
 $n_{base} = \lfloor k / |A| \rfloor, r = k \bmod |A|$

Sort active collections by min-distance ↑
distribute r to closest active sources

Global re-sort by distance ↑

Top-k docs

Filtered + balanced · full strategy

k-Fixing Constraint

Each strategy applied independently for $k \in \{3, 5, 10\}$
UMS@10 sliced to 5 ≠ UMS@5 directly

Ranked Evidence List → LLM Generator
Recall@k evaluated for each strategy independently